Can Language Models Learn a PDE? An Audit of Measurement Selection in Weak-Form Inverse Problems

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Abstract

Weak measurements of a fixed solution and its source provide a controlled setting for studying how language models choose informative observations. We study a one-dimensional elliptic boundary-value problem in which an agent selects test functions, a tool returns weighted source integrals, and a numerical solver fits the unknown coefficients by least squares. Our audit combines 180 archived scored runs from six models on 30 shared tasks with fixed and randomized query designs, numerical sensitivity tests, and validation on 300 fresh tasks. Fixed sine tests recover all 30 archive-derived tasks at the minimum permitted query count; on fresh tasks, success is 92%, 91%, and 98% across the three settings, including ten unidentifiable zero-forcing cases. An exact nonuniqueness result and a consistency–stability analysis distinguish ambiguity in the inverse problem from sensitivity of its discretization, while free diagnostic measurements complicate comparisons under a stored-query budget. These results delimit what coefficient-recovery scores establish about a language model's measurement-selection ability.

1. Introduction

A scientific agent must choose observations that distinguish plausible explanations. In an inverse problem, the question is concrete: which measurements identify the unknown parameters, and how reliably can those parameters be recovered? A controlled mathematical environment lets us study those choices while knowing the parameters used to generate the observations. It also makes the numerical assumptions behind an agent's score available for scrutiny.

Our setting is a one-dimensional elliptic boundary-value problem, representing a stationary diffusion–transport–reaction equation. The spatial response is available through tools, while the diffusion, transport, reaction, and source fields are hidden. The agent selects functions against which to integrate the equation. Each choice imposes another weak constraint on the same fixed system; the source and boundary conditions remain fixed throughout a task. Numerical tools assemble the resulting linear equations and fit the coefficient fields. The agent chooses what to measure, which diagnostics to request, and when to stop.

This setup connects two familiar questions. For PDE and inverse-problem readers, it asks whether weak measurements of one solution determine the operator and whether the discrete recovery is stable. For machine-learning readers, it asks whether a model's sequential choices improve a tool-assisted estimation task over simple fixed designs. The equation family and parameter count are disclosed, while the supplied solver handles the coefficient representation. "Learn" means identifying parameters within a session. The spatially one-dimensional problem is mathematically an ODE boundary-value problem, and does not test multidimensional PDE structure.

The benchmark first appeared in a Kaggle write-up (Miller, 2026). We reanalyze its six-model archive and ask: how do the recorded policies differ on shared tasks; what do simple numerical query designs achieve; and how do identifiability, numerical error, and the observation budget affect those comparisons? Our contributions are a precise account of the observation model, reproducible task-level comparisons, and mathematical and numerical controls. Fixed sine tests suffice on all 30 archive-derived tasks. Fresh-task validation and an exact ambiguity construction delimit that finding. The resulting case study shows why evaluation of scientific agents needs both an inverse-problem analysis and non-language-model controls.

2. Related Work

Interactive scientific agents. DiscoveryWorld evaluates a broad scientific-discovery loop in simulated worlds (Jansen et al., 2024). BoxingGym studies automated experimental design and model discovery across generative environments (Gandhi et al., 2025). SciGym provides a systems-biology

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dry lab for scientific-agent evaluation (Duan et al., 2025). Our task shares their sequential observation–feedback structure. Its narrower contribution is a finite-dimensional weak-form inverse problem in which the measurement matrix, coefficient truth, and numerical sensitivities can be audited directly. We do not claim to introduce interactive scientific evaluation itself.

Equation discovery and inverse problems. LLM-SR uses language models for scientific equation discovery, and LLM-SRBench evaluates discovery from observations (Shojaee et al., 2025a;b). Weak SINDy uses integral formulations for PDE identification (Messenger & Bortz, 2021). Here the operator family is fixed, and a supplied solver fits coefficients in a prescribed polynomial basis from the agent's chosen test functions. This is a measurement-selection problem within a specified model, rather than symbolic structure discovery. Optimal experimental design provides a relevant reference (MacKay, 1992; Alexanderian, 2021), but its criteria require an explicit noise model, admissible designs, and appropriate rank assumptions. We do not equate expression-family diversity with information gain.

Identifiability and confidence. Identifiability depends on the observation operator. Results for boundary maps, reaction–advection–diffusion processes, and parabolic observations (Egger et al., 2014; Browning et al., 2024; Salmaniw & Browning, 2025) do not by themselves guarantee uniqueness from one stationary solution and source. Our analysis distinguishes this structural issue from finite-grid conditioning. The confidence analysis uses an explicit coefficient-error event, following the distinction between stated confidence and empirical accuracy (Guo et al., 2017). The cognitive framework of Burnell et al. (2026) contextualizes the original behavioral labels; our diagnostics describe behavior under a specific prompted protocol.

3. Benchmark and Observation Model

3.1. Forward Problem, Observations, and Recovery

The hidden equation is

$$\begin{aligned} -(au')' + bu' + cu = f, \quad x \in (0,1), \\ u(0) = u(1) = 0. \end{aligned} \tag{1}$$

Here u is the spatial response, primes denote spatial derivatives, and a, b, c, f are diffusion, transport, reaction, and forcing, respectively. The forward problem computes u from these fields and boundary data. The inverse problem fixes that response and seeks the fields that produced it. The supplied generator and solver use the parameterization $q(x) = \sum_{j=0}^{N-1} \theta_{q,j} P_j(x)$, for $q \in \{a, b, c, f\}$, using ordinary shifted Legendre polynomials on $[0, 1]$. For example, $P_0(x) = 1$ and $P_1(x) = 2x - 1$. The fields are therefore polynomials of degree at most $N - 1$. This basis is orthogonal but not orthonormal.

At the continuum level, take smooth coefficients with positive diffusion and an observed solution $u \in H_0^1(0,1)$. Here H_0^1 consists of square-integrable functions with square-integrable first weak derivative and zero endpoint values. Multiplying Equation (1) by a test function $\varphi_k \in H_0^1(0,1)$ and integrating by parts gives

$$\begin{aligned} \int a u' \varphi_k' \, \mathrm{d}x + \int b u' \varphi_k \, \mathrm{d}x + \int c u \varphi_k \, \mathrm{d}x \\ = r_k, \qquad r_k = \int f \varphi_k \, \mathrm{d}x. \end{aligned} \tag{2}$$

All integrals are over $(0, 1)$. The endpoint condition removes the boundary term, so only first derivatives of u are needed. A query chooses the weighting function φ_k and observes the source integral r_k, also called a source moment. For example, $\varphi_k = \sin(k\pi x)$ requests a sine moment of the source.

Each test function supplies one linear equation in the unknown operator coefficients. Define

$$\begin{aligned} D_{kj} &= \int P_j u' \varphi_k' \, \mathrm{d}x, & B_{kj} &= \int P_j u' \varphi_k \, \mathrm{d}x, \\ C_{kj} &= \int P_j u \varphi_k \, \mathrm{d}x, & F_{kj} &= \int P_j \varphi_k \, \mathrm{d}x. \end{aligned} \tag{3}$$

After K queries, let $A = [D \mid B \mid C] \in \mathbb{R}^{K \times 3N}$ and $\theta = (\theta_a^\top, \theta_b^\top, \theta_c^\top)^\top$. The tool separately fits

$$\begin{aligned} \widehat{\theta} &\in \arg\min_{\theta} \|A\theta - r\|_2, \\ \widehat{\theta}_f &\in \arg\min_{\theta_f} \|F\theta_f - r\|_2. \end{aligned} \tag{4}$$

For a nonunique fit, the tool selects the minimum Euclidean norm solution under its numerical singular-value cutoff. Both fits use the measured vector r, so operator recovery does not require a preceding source estimate. Source moments provide data in addition to the observed solution; recovery is not based on u alone.

The agent chooses the test functions, rather than matrix entries. The tool constructs and stores each row internally; the agent receives r_k, then coefficient estimates and selected diagnostics when it requests a solve. The full joint matrices and spectra reported below are reconstructed for the audit. The tests define identification constraints; the forward solution is computed by finite differences, rather than expanded in these test functions. The protocol requires $K \geq 4N$ before a solve, although the operator fit has $3N$ unknowns. This rule does not guarantee rank or uniqueness.

Table 1. Session actions. Only accepted stored queries consume the query budget. Free checks reveal a source moment without adding a solver row.

Action	Returned information or effect
`eval_solution`	Values of u, u' at chosen points.
`QUERY`	r_k; appends rows to A, F.
`term_integrals`	The three term integrals before weighting by the coefficient basis.
`solve`	Least-squares coefficient estimates and changes.
`check`	Predicted weak left-hand side and actual source moment for a candidate expression.
`PREDICT`	Submits the latest solve coefficients.

Table 2. Reconstructed task settings. The archive records N, query and turn budgets. Generator parameters are verified by exact recovery of all 30 archived coefficient vectors. Sparsity is the generator's zeroing parameter; scale and basis dimension change together.

	Easy	Medium	Hard
N	2	3	4
Stored-query budget	12	18	24
Turn limit	36	54	72
Generator sparsity	.3	.2	.1
Generator scale	.5	1.0	1.5
Diffusion parameter $a_{\min}$	1.0	.5	.3
Solution grid points	2001	2001	4001

3.2. Actions, Information, and Budgets

Table 1 summarizes the interface. The recovered prompt implementation specifies the equation and parameter count but does not name the Legendre basis; its coordinate recovery is delegated to the solve tool. The model can evaluate the solution and its numerical derivative, choose symbolic expressions, inspect estimates, and request consistency checks. The parser admits mathematical expressions including polynomial, trigonometric, exponential, and localized functions, subject to sampled-value and endpoint checks. Numerical acceptance does not certify membership in the continuum test space; our predefined controls use smooth functions. There is no common amplitude normalization.

A turn is one model response, possibly containing reasoning and tool commands. The protocol permits at most one stored query per turn; several free commands can share a response. Submission cannot share a response with a query. The prompt encourages repeated solves and requests confidence reports. Therefore solve counts and iteration are behaviors under an instruction to iterate. A query budget counts stored rows, not all observations: a free check returns $\int f\varphi$ for a new expression, and solution evaluations are also free.

The generator checks diffusion positivity on 500 spatial points; the parameter $a_{\min}$ controls the constant diffusion coefficient, not a certified pointwise lower bound. Reaction can have either sign, so diffusion positivity alone does not guarantee a unique continuum forward solution. The evaluation takes the computed solution as data.

The reference implementation uses second-order finite differences for the solution, a fourth-order interior derivative with boundary stencils, and composite Simpson quadrature on the solution grid. It rounds each stored matrix entry and right-hand-side value to 12 decimal places. The solve uses unweighted least squares. The model sees coefficients rounded to eight decimal places; final scoring uses the internally retained unrounded estimate. Symbolic test functions are differentiated by a fourth-order centered stencil with step 10^{-6}, independently of the solution-grid spacing. These choices matter when singular values are small. Table 2 also shows that the three difficulty labels are bundled settings; differences cannot be attributed solely to dimension.

3.3. Coefficient Error, Uniqueness, and Stability

The score measures error in the prescribed coefficient coordinates:

$$E_q = \max_j |\widehat{\theta}_{q,j} - \theta_{q,j}|, \qquad E = \sum_{q\in\{a,b,c,f\}} E_q. \quad (5)$$

The primary success event is $E < 0.01$; $E < 0.1$ is a sensitivity check. This absolute, basis-dependent parameter error is distinct from a PDE residual or a function error. For each field, $\|\widehat{q} - q\|_{L^\infty(0,1)} \le NE_q$, since $|P_j| \le 1$. Thus a small coefficient error controls field error here, but the same numerical thresholds for the two quantities are not equivalent.

Three questions should be separated. *Structural identifiability* asks whether different coefficient triples can produce the same u, f, even with all admissible weak measurements. *Measurement selection* asks whether a finite set of tests distinguishes triples in the chosen coefficient space. *Numerical stability* asks how small data and discretization errors affect the recovered coefficients. Full column rank of an exact moment matrix formed from the continuum solution would certify uniqueness in the operator coefficient space. Full rank after numerical approximation and rounding need not do so. Separate full-rank blocks D, B, C do not exclude dependencies between blocks.

To see the role of numerical error, write A_h, r_h for the assembled system and θ_* for the true operator coefficients. Its consistency defect is $e_h = A_h\theta_* - r_h$. The forward solve, derivative approximation, quadrature, and rounding need not preserve the weak identity exactly, so this defect can be nonzero even at the truth. If A_h has full column rank,

exact least squares for that assembled system gives

$$\widehat{\theta} - \theta_* = -A_h^{\dagger} e_h,$$
$$\|\widehat{\theta} - \theta_*\|_2 \le \frac{\|e_h\|_2}{\sigma_{\min}(A_h)}. \tag{6}$$

Here $A_h^{\dagger}$ is the pseudoinverse. The bound concerns the operator fit; the score also includes the separately fitted source. A small fitted weak residual $\|A_h\widehat{\theta} - r_h\|_2$ need not imply a small coefficient error: a small singular value can amplify a small consistency defect. The forward finite-difference residual, fitted weak residual, and coefficient score therefore answer different questions.

Let $\mathcal{P}_d$ denote polynomials of degree at most d. The following example shows an ambiguity that persists even before discretization.

Proposition 1 (An admissible nonidentifiable family)**.** *Let $u(x) = x(1-x)$, $N \ge 2$, and $T(\delta a, \delta b, \delta c) = -(\delta a u')' + \delta b u' + \delta c u$ with domain $\mathcal{P}^3_{N-1}$. Then T has rank $N+2$ and nullity $2N-2$. Every element of its kernel leaves all continuum weak-form measurements unchanged at fixed u, f.*

Even arbitrarily many exact weak measurements cannot distinguish operators separated by these kernel directions. Appendix A gives the proof and examples with positive diffusion and nonnegative reaction. This quadratic- solution family is an admissible counterexample, not a classification of the 30 sampled instances. There is also a positive-probability zero-excitation event in the generator: each forcing coefficient is zeroed independently with probability s, giving $\Pr(f \equiv 0) = s^N$, or 0.09, 0.008, and 0.0001 in the three settings. When $f \equiv 0$, under homogeneous boundary data the native forward solve returns $u = 0$ when its matrix is nonsingular. Then every operator design row vanishes, irrespective of the test function, so no measurement policy can identify a, b, c from these observations. The nonzero-source construction in Proposition 1 is a separate ambiguity mechanism.

4. Evaluation Design

4.1. Archive and Provenance

A *task* is one generated coefficient tuple and its numerical solution; a *run* is one agent's interaction with a task. A *policy* is a rule for selecting actions, fixed in advance or dependent on feedback. A random *seed* initializes the reproducible task generator. Thus six models on 30 shared tasks give 180 runs, not 180 different equations.

The public Kaggle dataset, *Elliptic Differential Equation LLM Reasoning*, version 1, contains 180 scored logs: six models, three difficulty settings, and seeds 1–10 in every cell. The same 30 ground-truth coefficient tuples are shared across models. The archive was released on 16 April 2026; the provider model IDs are listed in Appendix B. All logs record the `standard` prompt, called enhanced in the write-up. The archive does not contain a baseline-prompt ablation, repeated stochastic draws per model–task pair, or a complete record of earlier failed attempts and retries. Rates are consequently conditional on inclusion in the scored archive.

We verify the archive checksum, cell membership, shared truths, finite scores, and agreement with the saved sweep summary before analysis. The exact source revision and temperature are not recorded per run. A public implementation snapshot postdates some runs. Our new experiments use the current pinned numerical environment; they do not recreate historical provider calls.

4.2. Numerical Controls and Replay

For the controls, we regenerate all 30 tasks from their recorded seeds, verify exact equality with saved ground-truth coefficient vectors, and evaluate non-language-model query policies under the same accepted-query limits and supplied query/solve interface. The policies use no hidden coefficients, solution values, measurement responses, or diagnostics to select expressions. Fixed sine uses the Dirichlet Laplacian eigenfunctions $\varphi_k = \sin(k\pi x)$, $k = 1, \dots, K_{\max}$, where $K_{\max} = 6N$ is the stored-query budget. Boundary Legendre multiplies a polynomial test by the endpoint-vanishing factor $x(1-x)$, giving $x(1-x)P_{k-1}(x)$. The implementation uses factored expressions for numerical stability. Random mixed draws uniformly from five admissible families, using ten independently seeded query lists per setting, reused across the ten PDE seeds. Its fixed parameter ranges and exact expressions are released in the experiment output. All controls solve at every query count from $4N$ through $6N$; additional solves do not alter their predefined query lists. These controls test whether simple nonadaptive policies suffice on these instances; they do not reproduce the models' free-action or stopping decisions.

For the numerical audit, we replay the archived query expressions, compare the returned source moments with the logs, and reconstruct joint matrix spectra. We report both raw and unit-column-norm spectra, with explicit relative singular value cutoffs rather than a binary rank label alone. We additionally multiply each reconstructed row and its source moment by a common amplitude before 12-decimal rounding, leaving the query sets fixed and refitting all stored rows. This audit scores 8-decimal diagnostic estimates, unlike unrounded primary final scores. In exact arithmetic this common scaling leaves least squares unchanged; its effect here tests the absolute quantization rule. It is a post hoc numerical sensitivity analysis, rather than an optimized query policy or a model rerun.

We also refine the forward and quadrature grids together by

factors two and four. Original-resolution primary controls use the native dense solver. Refinements use an independently assembled banded solver for the same finite- difference equations; we validate its solution and score sensitivity against native solves on all 30 tasks. Refinement changes the numerical environment, rather than adding independent tasks or measuring observation-noise robustness.

To test specificity to the archived seeds, we additionally evaluate the fixed sine policy on 100 fresh seeds (1001–1100) in each difficulty, with the original generator and grid settings. Every planned task is retained, including zero forcing and numerical failures; there are no retries or filters. This is validation after design selection, not a preregistered comparison or a new language-model experiment. We report outcomes on all 100 tasks per setting, with Wilson intervals and descriptive zero/nonzero-source breakdowns. Reused seed labels across difficulties are not pooled into independent sampling units.

4.3. Task-Level Comparisons and Confidence

The comparison unit is the shared PDE seed within each difficulty, not an individual coefficient, solve, or confidence report. Each model has ten tasks per setting. We report Wilson 95% intervals for marginal success proportions and paired task resampling for success differences (20,000 draws; seed 20261020). Exploratory pairwise tests condition on discordant tasks, with Holm correction across the 15 model pairs within each difficulty. There are no repeated model draws from which to estimate within-task response variance. Empirical paired-bootstrap intervals can degenerate when all ten observed task differences agree. These intervals describe uncertainty under task resampling assumptions and do not resolve selection into the scored archive.

A confidence report states the model's probability that a coefficient block is accurate. We pair it only with a solve returned on a strictly earlier turn. Its correctness event is $E_q < 0.1$, separately for $q \in \{a, b, c\}$, which differs from final-run success $E < 0.01$. All reports and coefficient blocks within a run are kept together when resampling; run-balanced scores avoid giving a model more weight merely because it reports more often. Action counts, first-to-last solve improvement, and expression-family entropy are descriptive diagnostics. They do not independently validate planning, learning, or information gain.

5. Results

5.1. Archived Recovery Differs, with Limited Resolution

Figure 1 shows the archived success counts. At $E < 0.01$, Gemini 3.1 Pro succeeds on $10/10$, $9/10$, and $10/10$ easy, medium, and hard tasks; Claude Opus 4.6 achieves $10/10$, $9/10$, and $8/10$. GPT-5.4 achieves $7/10$, $3/10$, and $1/10$.

Table 3. New offline controls at the original resolution and full stored-query budget. Deterministic policies have ten tasks per difficulty. Random rows summarize ten query-list replicates on the same ten tasks; the denominator 100 is not 100 independent PDEs.

Policy	Setting	Successes	Median E
Fixed sine	Easy	10/10	9.5e-05
Fixed sine	Medium	10/10	0.0013
Fixed sine	Hard	10/10	0.00057
Boundary Legendre	Easy	10/10	3.4e-05
Boundary Legendre	Medium	9/10	0.0014
Boundary Legendre	Hard	10/10	0.00057
Random mixed	Easy	100/100	6.4e-05
Random mixed	Medium	92/100	0.0016
Random mixed	Hard	85/100	0.00097

The other models also vary substantially across settings and seeds. Corrections to the earlier write-up figures are documented in Appendix B.

With ten tasks, even $10/10$ has a Wilson 95% interval of approximately $[0.72, 1.00]$, and $9/10$ has $[0.60, 0.98]$. Thus the heatmap is not a precise model ranking. All pairwise task-level comparisons and confidence intervals are released with the analysis; differences in this selected archive should not be generalized to end-to-end provider reliability. Only one of 45 exploratory pairwise comparisons has adjusted $p < 0.05$ under the stated within-difficulty Holm procedure: Gemini 3.1 Pro versus Claude Sonnet 4 on hard tasks ($10/10$ versus $0/10$, adjusted $p = 0.0293$). There is no correction across the three difficulty families; this finding does not establish a global model ordering.

5.2. Simple Nonadaptive Queries Suffice on These Tasks

The fixed sine policy reaches $E < 0.01$ on all ten tasks in each difficulty, already at the minimum $4N$ stored queries (8, 12, and 16). It retains 30/30 success at the full $6N$ budget. Boundary Legendre succeeds on 29/30 at full budget. Random mixed succeeds on 100/100 easy, 92/100 medium, and 85/100 hard policy–task combinations (Table 3); these reuse ten PDE instances per setting and are not 100 independent tasks. Across random query-list replicates, hard success ranges from 6/10 to 10/10. Figure 2 shows recovery at every permitted solve count.

These controls show that the archived tasks can be solved by a simple nonadaptive design with the supplied numerical tools. They do not show a statistically established advantage over the best archived model, nor isolate reasoning from parsing and stopping behavior. Because exact historical internal matrices were not retained, we compare newly reconstructed controls with archived outcomes transparently rather than claiming a prospective provider comparison. The evidence does not support a claim that language-model

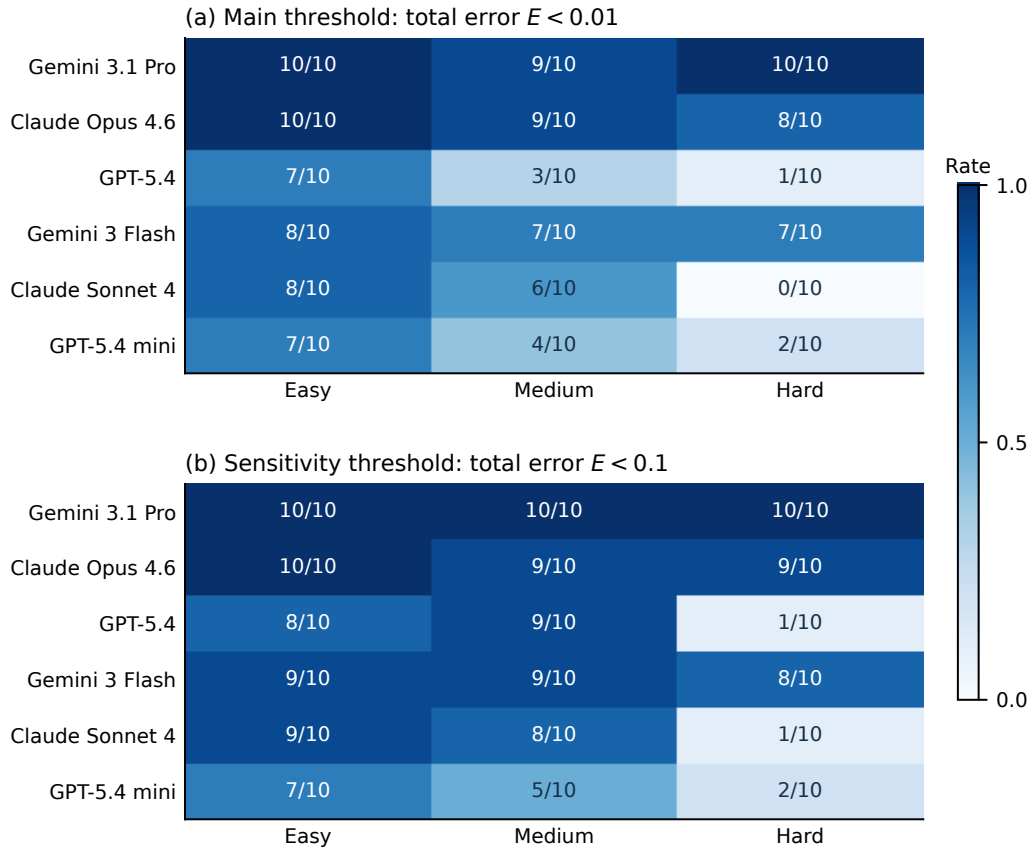


Figure 1. Archived recovery on 30 shared tasks, with ten scored records per model and difficulty. The two panels use total maximum-coefficient-error thresholds $E < 0.01$ and $E < 0.1$. Counts are conditional on the scored archive; unscored attempts and retries are not represented. The primary threshold was retained from the legacy success-plot convention; the scorer itself returns continuous errors.

adaptation is necessary for recovery on this task set.

Grid refinement improves numerical errors: for fixed sine at full budget, median E changes from 9.47×10^{-5} to 2.18×10^{-6} on easy, 1.32×10^{-3} to 2.82×10^{-5} on medium, and 5.65×10^{-4} to 1.56×10^{-5} on hard when the grid is refined fourfold. At twofold refinement, random mixed reaches 100/100 hard successes and boundary Legendre reaches 10/10 medium successes. The original-grid native and banded solves agree on both success thresholds at every checkpoint across all 360 paired policy–task runs. Thus the banded implementation does not change these threshold comparisons, although individual numerical errors differ.

5.3. Fresh Seeds Expose the Limits of the Archive Result

Table 4 validates the fixed sine policy on all 300 fresh tasks. At the minimum query budget, success is 92/100 easy, 91/100 medium, and 98/100 hard. At full budget, it is 92/100, 85/100, and 93/100, respectively. All 300 executions complete without errors; no task is retried or excluded.

Table 4. Fixed sine on 100 fresh seeds per setting. All planned tasks remain in the denominator. Parentheses give Wilson 95% intervals for the minimum-budget rate; zero-source counts are descriptive.

Setting	$4N$ successes	$6N$ successes	$f = 0$
Easy	92/100 (0.85, 0.96)	92/100	7
Medium	91/100 (0.84, 0.95)	85/100	3
Hard	98/100 (0.93, 0.99)	93/100	0

Seven easy and three medium tasks have zero forcing, exactly zero solutions, and joint operator rank zero at both budgets. All ten fail, as predicted by the zero-excitation argument. Excluding them only for a descriptive breakdown, full-budget recovery is 92/93 easy, 85/97 medium, and 93/100 hard.

Thus the archived 30/30 result does not establish perfect

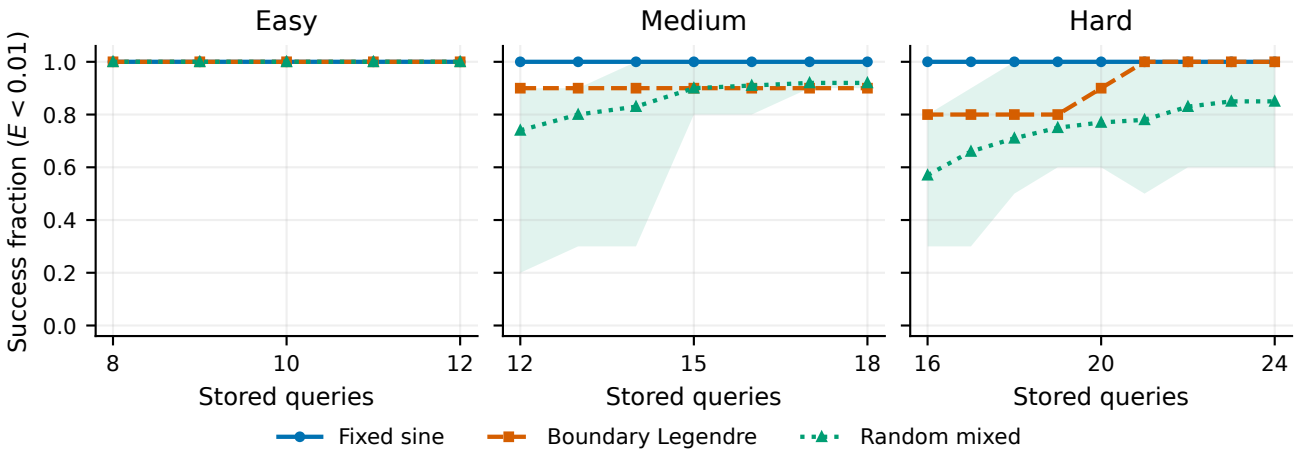


Figure 2. New controls on the 30 reconstructed tasks at the original grid resolution. Fixed sine and boundary Legendre use one predefined query list per setting. Random mixed averages ten independently seeded lists, each reused on the same ten tasks; shading is the minimum–maximum success fraction across lists, not a confidence interval. No query policy uses feedback or hidden truth.

generator-wide recovery. Nonzero-source failures also remain, and increasing the stored-query budget can reduce the number of coefficient-threshold successes in the same fixed policy. More rows do not guarantee lower coefficient error in an inconsistent finite-precision system. This validation evaluates one selected numerical policy; it does not determine whether adaptive policies improve recovery on fresh tasks. It does establish why a benchmark needs both simple controls and a stated treatment of unidentifiable instances.

5.4. What Numerical Reconstruction Can Establish

All 30 ground-truth coefficient vectors regenerate exactly, and the reconstructed reference implementation reproduces all 2,874 archived query source moments exactly at the stored precision. Source-moment agreement does not test the historical forward solution or unretained operator matrix: the source moment depends only on f, the test function, and quadrature. It does not establish exact historical solve replay: across 687 solves, the median maximum deviation from the logged 8-decimal coefficient vectors is 2.55×10^{-5}, and the largest is 3.50. The archive lacks historical internal matrices and sufficient environment provenance to localize every discrepancy. New controls and spectral diagnostics must therefore be identified as reconstructions.

All 180 reconstructed final stored-query matrices have full column rank at NumPy's default floating-point tolerance. After unit-column scaling, however, the median $\sigma_{\min}/\sigma_{\max}$ is 2.26×10^{-7}, 5.53×10^{-10}, and 1.80×10^{-10} for easy, medium, and hard runs. At a declared relative cutoff of 10^{-8}, 13/60, 42/60, and 37/60 matrices are rank deficient. These are diagnostics of limited numerical resolution, not continuum nonuniqueness certificates for the sampled instances.

Common-amplitude scaling changes the reconstruction's $E < 0.01$ counts (Figure 3). At amplitudes $10^{-3}, 1, 10^{3}$, the counts are respectively 24/50/53 of 60 easy runs, 11/36/48 medium runs, and 19/28/33 hard runs. No additional test functions are selected: the exact-arithmetic measurements differ only by a common nonzero factor, while their relative quantization precision changes. The changed counts show that absolute rounding and ill conditioning can alter a success event even when the underlying span of measurements is unchanged. They do not establish that larger amplitude universally improves recovery.

5.5. Information Use and Confidence

The archive contains 459 free consistency checks; 457 check expressions are absent as literal strings from earlier stored-query histories. Literal novelty does not prove semantic independence, but each check exposes an actual source moment. Thus stored-query counts alone do not equate the information acquired by two policies. This behavior is permitted by the protocol.

Appendix C reports confidence, action use, improvement between solves, and expression-family diversity. Confidence reports are matched to the estimates available when they were made. Repeated reports remain grouped within their run, and expression diversity is not equated with independent constraints. These diagnostics describe interaction behavior under the supplied prompts.

6. Discussion and Limitations

The audit reveals three requirements for interpreting measurement-selection benchmarks. First, the observation model must distinguish identifiable parameters from equiv-

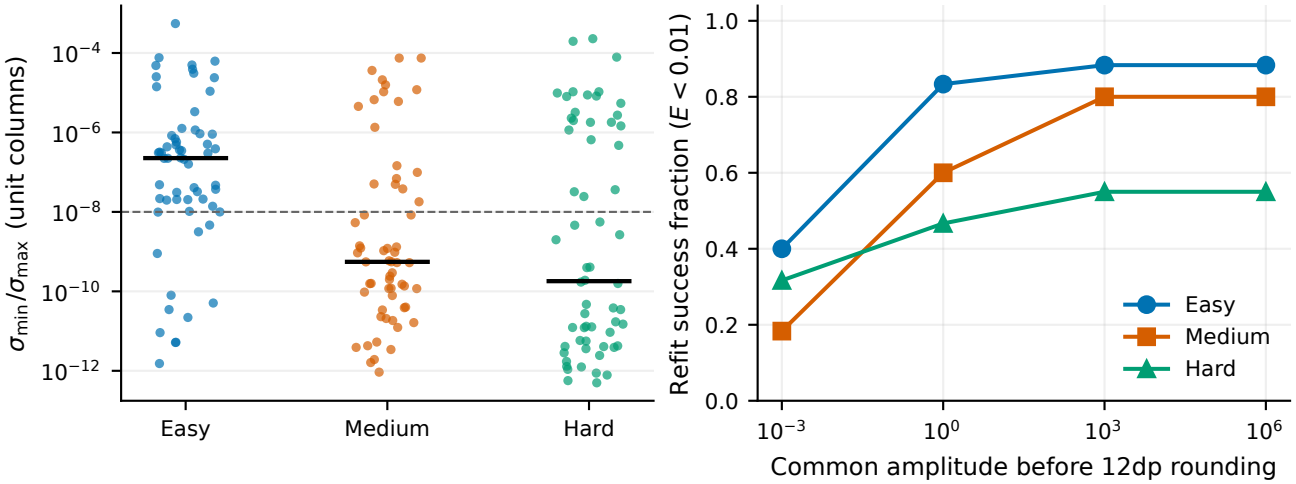


Figure 3. Numerical audit of reconstructed archived measurements. Left: unit-column-norm smallest-to-largest singular-value ratios, one point per model–task query design. The dashed line is a diagnostic cutoff, not a theorem of nonidentifiability. Right: strict-threshold success after common row/RHS scaling before absolute 12-decimal rounding, using all stored queries and fixed archived expression choices. These post hoc solves use reconstructed matrices and 8-decimal diagnostic estimates, and include later stored queries when a run stopped solving before its final query; they are not archived final submissions.

alent operators. A finite matrix's numerical rank does not replace this analysis. Second, normalization, precision, and discretization must be specified and tested. Equivalent continuum measurement designs can have different rounding errors; that distinction should be explicit in claims about the ability being measured. Third, every truth-dependent observation must enter the information accounting, including diagnostics exposed as free tools.

Our evidence is deliberately bounded. The model archive contains 30 tasks and one scored response trajectory per model–task pair, rather than a prospective all-attempt study. Difficulty settings conflate dimension, scale, sparsity, and resolution. We cannot reconstruct missing attempt histories or exact provider settings. The structural proposition concerns an admissible subclass and does not determine the continuum rank of every sampled operator. Reconstructed spectra, amplitude tests, and numerical controls inherit the current solver and must not be presented as historical model reruns.

A prospective benchmark should preregister the task distribution, retain every attempt and failure, fix model versions and generation settings, and compare fixed, randomized, and adaptive designs under a common observation budget. It should normalize admissible tests, specify a noise or precision model, characterize joint observability, and include prediction under independent sources or boundary data where coefficient uniqueness is required. Confidence scores should use a defined event and a declared reporting schedule. These are requirements for a stronger experimental claim, rather than quantities that can be inferred from the present archive.

7. Conclusion

Weak-form inverse problems make parts of scientific-agent evaluation directly inspectable. The archived models differ in recovery and interaction behavior, but an interpretation in terms of experimental-design ability requires controls for the observation model, numerical conditioning, and information accounting. A fixed sine design suffices on all 30 archive-derived tasks; fresh seeds reveal unidentifiable zero-excitation cases and additional coefficient-recovery failures. Both results make numerical baselines essential to interpreting model scores. The released analysis provides checks for this benchmark and a concrete basis for designing a more controlled prospective study.

Impact Statement

This work studies a small mathematical environment for evaluating scientific agents. Its immediate use is methodological: making evaluation claims traceable to observations, numerical assumptions, and complete records. Overinterpreting such scores could distort comparisons of scientific capabilities; the audit limits that risk by documenting what the measurements support. We do not infer readiness for autonomous scientific deployment from these experiments.

Reproducibility

Code, pinned data provenance, and analysis scripts are available at `https://github.com/Hydrodynamical/PDE_Learning`. Appendix B gives data and numerical details. The offline analyses require no new model calls.

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A. Structural Nonuniqueness: Proof and Examples

Proof of Proposition 1. Let $\mathcal{P}_d$ denote polynomials of degree at most d. For $u = x(1-x)$, all three terms in T lie in $\mathcal{P}_{N+1}$, so the rank is at most $N+2$. The reaction component contains $u\mathcal{P}_{N-1}$. Taking constant δa gives a nonzero constant because $-u'' = 2$, and taking constant δb gives $u' = 1 - 2x$. These span $\mathcal{P}_1$. Polynomial division by the quadratic u gives $\mathcal{P}_{N+1} = u\mathcal{P}_{N-1} \oplus \mathcal{P}_1$, proving surjectivity and rank $N+2$. The domain has dimension $3N$, so rank–nullity gives $\dim\ker T = 2N - 2$. Multiplication by any admissible test function and integration by parts shows that a kernel element changes no weak measurement. □

For $N = 2$, $u = x(1-x)$ and $f = 3$, the operators

$$a = \tfrac{3}{2} + \varepsilon, \qquad b = \varepsilon(4x-2), \qquad c = -8\varepsilon$$

all satisfy the same equation, since $2\varepsilon + \varepsilon(4x-2)(1-2x) - 8\varepsilon x(1-x) = 0$. For example $\varepsilon = -1/8$ preserves positive diffusion and gives nonnegative reaction. For $N \geq 3$, the base choice $a = 1, b = 0, c = 1, f = 2 + u$ belongs to the class. Sufficiently small perturbations in every kernel direction preserve positive diffusion and reaction, giving a relatively open $(2N-2)$-dimensional family in the affine kernel of admissible coefficient tuples with identical observations. Neither construction is asserted to occur among the 30 sampled tasks. The audit script also verifies ranks over exact rational arithmetic for $N = 2, \ldots, 8$; these checks supplement the proof rather than replace it.

B. Data, Implementation, and Statistical Details

The primary input is the public Kaggle dataset *Elliptic Differential Equation LLM Reasoning*, version 1. The repository's `paper_artifacts/data_manifest.json` records the dataset URL, version, SHA-256 digest, expected members, model IDs, seeds, and logged settings. Analysis scripts validate the pinned archive before reading its JSON members; they do not execute archive contents. The reference environment is Python 3.12 with pinned numerical packages in `requirements.txt`. Derived JSON files retain per-task observations, method definitions, and numerical cutoffs.

Table 5. Exact provider IDs in the April 2026 archive. Display names describe that archived experiment, rather than current provider offerings.

Display name	Recorded model ID
Gemini 3.1 Pro	`gemini-3.1-pro-preview`
Claude Opus 4.6	`claude-opus-4-6`
GPT-5.4	`gpt-5.4`
Gemini 3 Flash	`gemini-3-flash-preview`
Claude Sonnet 4	`claude-sonnet-4-20250514`
GPT-5.4 mini	`gpt-5.4-mini`

OpenAI records specify `reasoning_effort=medium`; the other providers have no recorded override. Temperature is absent. The *PDE Library*, version 9, is a separately published source snapshot that postdates some runs, not an exact per-run revision. Exact coefficient regeneration and query-RHS agreement narrow the reproduction gap, but the solve discrepancies reported in the main text prohibit a claim of bitwise historical replay.

The original write-up describes basis sizes 3/5/8 and a held-out-grid function success criterion; the logs and scoring code establish 2/3/4 and Equation (5). The regenerated strict-threshold figure also corrects the hard GPT-5.4 mini count from 0/10 to 2/10, matching the individual logs and saved sweep summary. These corrections are recorded in the derived summaries. The earlier draft's distinct three-model, two-seed pilot lacks source logs and used an unverified error norm. Its numeric tables are preserved in `paper_artifacts/legacy_pilot_tables.tex` for provenance, but excluded from this manuscript's empirical evidence.

Table 6. Strict-threshold archived successes with Wilson 95% intervals. Each cell uses ten scored tasks; intervals cannot account for missing attempts or model-sampling variability.

Model	Easy	Medium	Hard
Gemini 3.1 Pro	10/10 [0.72, 1.00]	9/10 [0.60, 0.98]	10/10 [0.72, 1.00]
Claude Opus 4.6	10/10 [0.72, 1.00]	9/10 [0.60, 0.98]	8/10 [0.49, 0.94]
GPT-5.4	7/10 [0.40, 0.89]	3/10 [0.11, 0.60]	1/10 [0.02, 0.40]
Gemini 3 Flash	8/10 [0.49, 0.94]	7/10 [0.40, 0.89]	7/10 [0.40, 0.89]
Claude Sonnet 4	8/10 [0.49, 0.94]	6/10 [0.31, 0.83]	0/10 [0.00, 0.28]
GPT-5.4 mini	7/10 [0.40, 0.89]	4/10 [0.17, 0.69]	2/10 [0.06, 0.51]

Table 7. Exploratory exact paired success tests, Holm-adjusted within the 15 model pairs of each difficulty. All 45 comparisons are shown. Values are conditional on the retained scored tasks; no adjustment is made across difficulties.

First model	Second model	Easy	Medium	Hard
Gemini 3.1 Pro	Claude Opus 4.6	1.0000	1.0000	1.0000
Gemini 3.1 Pro	GPT-5.4	1.0000	0.9141	0.0547
Gemini 3.1 Pro	Gemini 3 Flash	1.0000	1.0000	1.0000
Gemini 3.1 Pro	Claude Sonnet 4	1.0000	1.0000	0.0293
Gemini 3.1 Pro	GPT-5.4 mini	1.0000	0.8750	0.1016
Claude Opus 4.6	GPT-5.4	1.0000	0.4688	0.1719
Claude Opus 4.6	Gemini 3 Flash	1.0000	1.0000	1.0000
Claude Opus 4.6	Claude Sonnet 4	1.0000	1.0000	0.1016
Claude Opus 4.6	GPT-5.4 mini	1.0000	1.0000	0.2812
GPT-5.4	Gemini 3 Flash	1.0000	1.0000	0.5000
GPT-5.4	Claude Sonnet 4	1.0000	1.0000	1.0000
GPT-5.4	GPT-5.4 mini	1.0000	1.0000	1.0000
Gemini 3 Flash	Claude Sonnet 4	1.0000	1.0000	0.1719
Gemini 3 Flash	GPT-5.4 mini	1.0000	1.0000	0.5000
Claude Sonnet 4	GPT-5.4 mini	1.0000	1.0000	1.0000

C. Behavioral Figures and Their Interpretation

The figures below regenerate the relevant Kaggle diagnostics from the full archive. They retain the original operational questions while correcting data or plotting discrepancies. For action counts, stacked marginal medians need not sum to the median total action count. For improvement, $R = E_{\text{first}}/E_{\text{last}}$; missing ratios, including single-solve runs, receive the historical display default $R = 1$. Large ratios can reflect poor initial estimates or the ordinary benefit of additional data.

Query-family entropy is $H = -\sum_{i=1}^{5} p_i \log_2 p_i / \log_2 5$, with $0 \log 0 = 0$. The five categories are polynomial, trigonometric, exponential, localized, and other, assigned by a heuristic string classifier. The archived taxonomy is recorded at the final solve; later queries in six runs are absent from it. A high entropy does not imply linearly independent or useful measurements.

The confidence plots use disjoint bins $[l, h)$, with 1 included in the last bin, and show mean confidence against empirical correctness. Reports are matched only to strictly earlier solves, since a solve requested in a report's own response has not yet been returned. The archive contains 733 confidence reports: 707 have a preceding solve and 26 are excluded. Causal matching changes the solve paired to 71 reports relative to the legacy query-count rule; 827 observations lie on interior bin boundaries that the old plots could count twice. The retained reports contribute 2,121 dependent observations for a, b, c, excluding f. Individual coefficient reports are not treated as independent tasks.

For run i with m_i retained coefficient observations, the run-balanced Brier score is

$$\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{r=1}^{m_i}(p_{ir} - y_{ir})^2.$$

We also report the signed gap obtained by replacing the square with $p_{ir} - y_{ir}$; positive values indicate average overconfidence. This gap is calibration in the large, not expected calibration error or a full calibration criterion. Intervals resample whole supported runs with 20,000 draws. There are 179 supported runs: Gemini Flash hard has nine and all other cells ten. Run balancing does not align reporting stages, query counts, or base accuracies. Thus these scores cannot compare intrinsic metacognitive ability without a matched reporting schedule and correctness distribution.

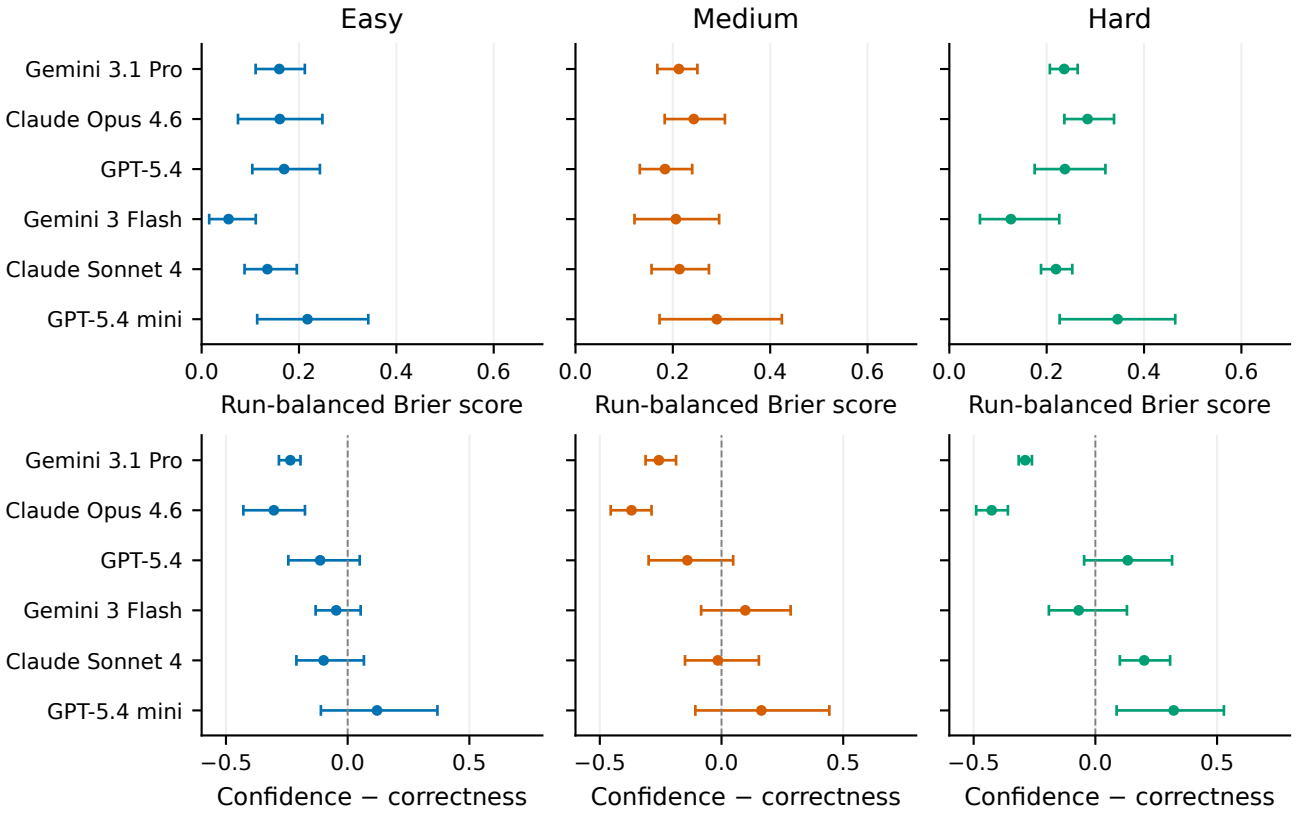


Figure 4. Run-balanced confidence summaries with whole-run percentile-bootstrap 95% intervals, conditional on having a preceding solve and confidence report. Brier scores reflect both correctness and stated confidence. Signed gaps measure average confidence minus correctness; zero alone does not establish calibration. Reports and coefficient blocks remain clustered within runs.

D. Why Legacy Sensitivity Scores Are Not Uncertainty Estimates

The archived code forms a truncated pseudoinverse H_k of $A_k^\top A_k$, discarding singular values below 10^{-10} times the largest. It summarizes each coefficient block using $s_k(q) = \sqrt{\sum_{i \in q} (H_k)_{ii}/N}$. A truncated inverse can assign zero weight to unresolved directions. Without an observation-noise model and rank conditions, it is not a calibrated coefficient covariance. Separate block values also cannot diagnose dependencies between blocks.

The monitoring routine ranks values using double `argsort`, without averaging ties; it is not standard tie-aware Spearman correlation. The control routine applies a Sherman–Morrison-style update to a truncated pseudoinverse, which need not agree with a freshly recomputed truncated inverse, and compares reductions across blocks for the chosen next row rather than across candidate queries. We retain these source fields for provenance but do not use them as evidence of calibrated uncertainty or optimal experimental design.

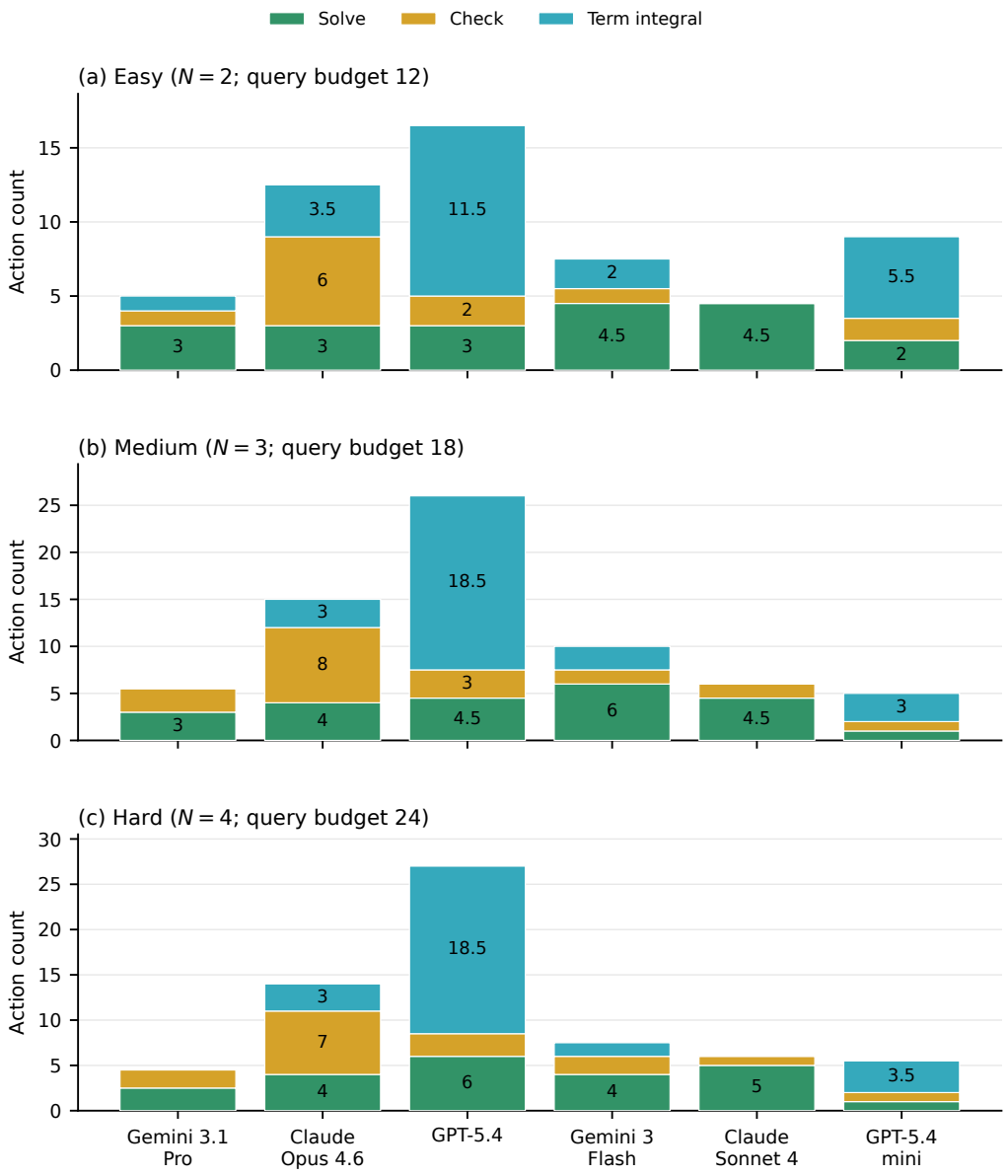


Figure 5. Archived tool use, with one panel per difficulty. Stacked components are separate marginal medians across ten runs per model. The prompt encourages iteration and checks supply source moments, so more actions do not establish better experimental design.

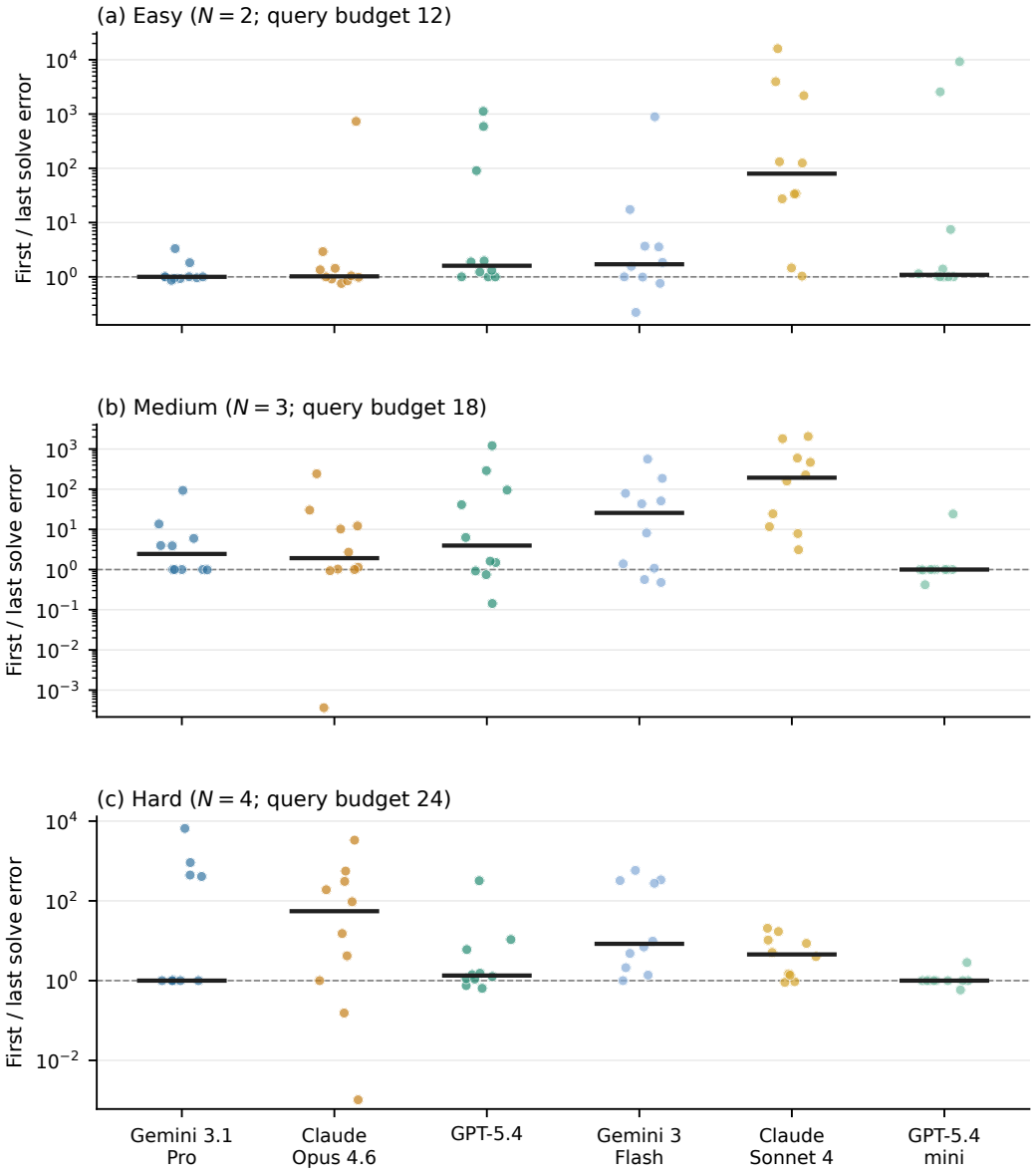


Figure 6. First-to-last solve improvement on a logarithmic scale. Points are runs and black markers are medians. Missing ratios have display value one. This statistic measures within-session error change, not model-weight learning.

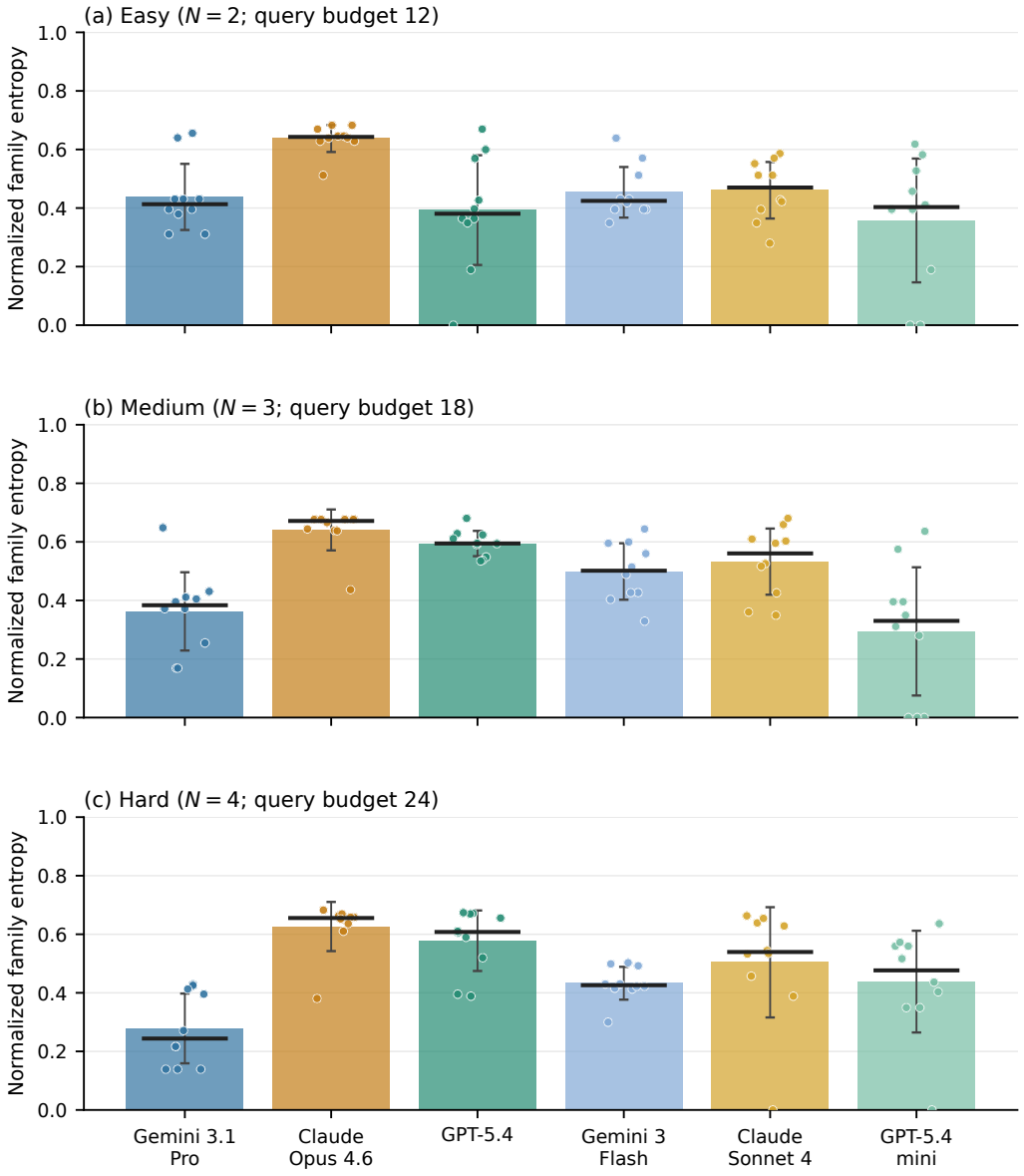


Figure 7. Normalized expression-family entropy. Bars are means, whiskers one population standard deviation, points individual runs, and black markers medians. The taxonomy describes expression forms, not measurement independence.

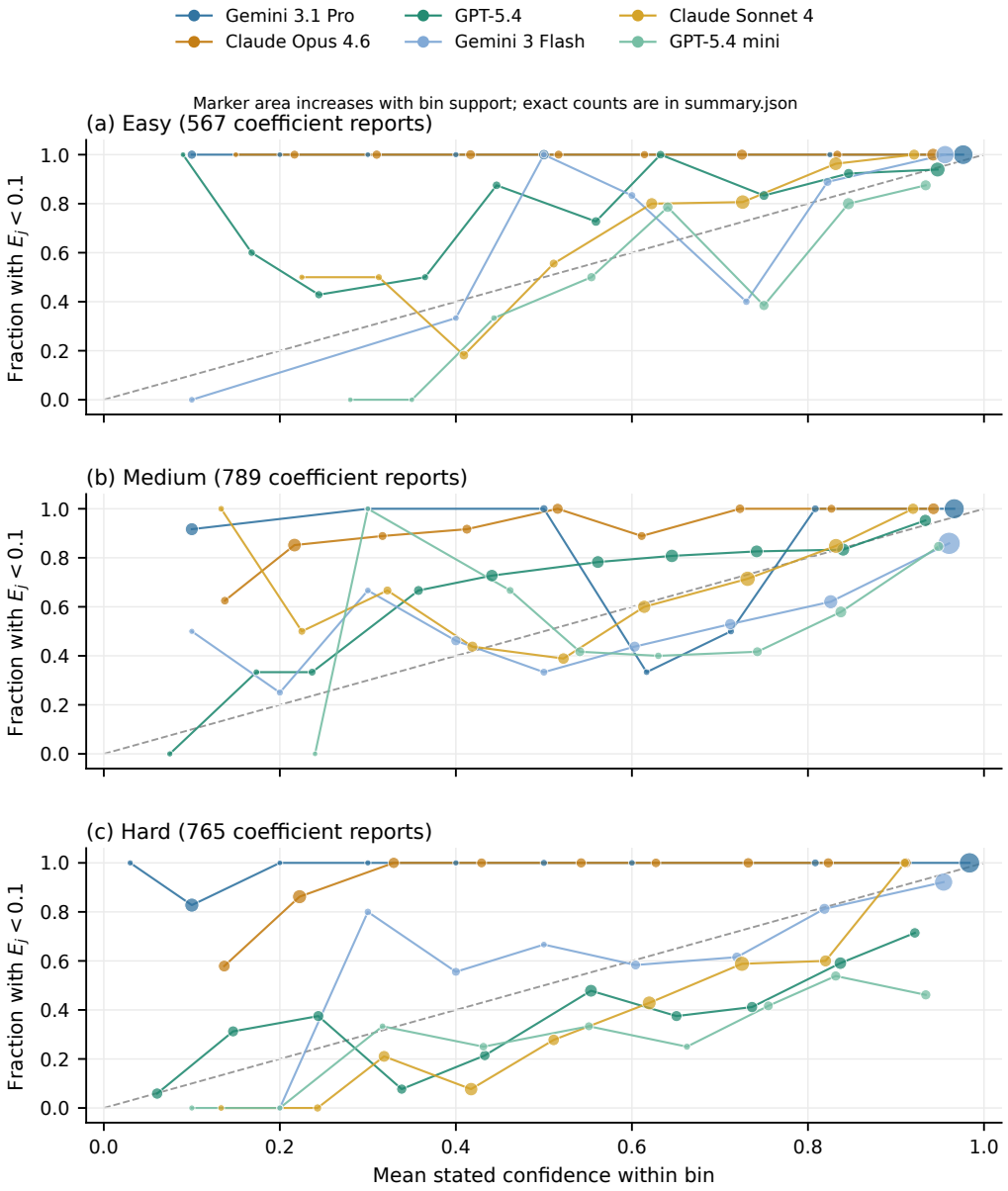


Figure 8. Corrected descriptive confidence curves. The horizontal coordinate is mean stated confidence in a disjoint width-0.1 bin; the vertical coordinate is the fraction with $E_q < 0.1$. Marker area grows with bin support. The 707 causally paired reports contribute 2,121 observations, clustered within runs; these curves alone are not a general calibration ranking.